AP & RAP

Quiz 1

Date: 11 November 2025
Time allowed: 60 minutes **Full mark: 100**

1. (Definitions and examples, 20 points)
   (a) Provide the definition of a measure; give two examples.
   (b) Provide the definition of a measurable function; give two examples.
   (c) Provide the definition of a simple function on a measurable space.

2. (Short questions, 25 points) In the following questions, no justification is necessary. Simply provide the shortest possible correct response. Each question is worth 5 points.
   (a) Let f be a measurable function on a measurable space. Can we always find a sequenc of simple functions that converge pointwise to f?
   (b) Is the pointwise limit of measurable functions always measurable?
   (c) Is the intersection of two σ-algebras again a σ-algebra?
   (d) Suppose f is an integrable function on a measure space $(\Omega, \mathcal{A}, \mu)$. Is it true that $\int_A f d\mu$ is finite for every $A \in \mathcal{A}$?
   (e) Suppose f is a nonnegative measurable function on a measure space $(\Omega, \mathcal{A}, \mu)$ and $\int_\Omega f d\mu = 0$. Can we conclude that $f = 0$ a.e.?

3. (Theorems and proofs, 20 points) Please state the dominated convergence theorem and prove it.

4. (20 points) Let f and g be extended real-valued measurable functions on $(\Omega, \mathcal{A})$, and define

$$h(\omega) = \begin{cases} f(\omega) & \text{if} \quad \omega \in A \\ g(\omega) & \text{if} \quad \omega \in A^c \end{cases}$$

   where A is a set in $\mathcal{A}$. Show that h is measurable.

5. (15 points) Let $(\Omega, \mathcal{A}, \mu)$ be a measure space.
   (a) If $A_n, A \in \mathcal{A}$ and $A_n \uparrow A$, show that $\mu(A_n) \uparrow \mu(A)$.
   (b) If $A_n, A \in \mathcal{A}$ and $A_n \downarrow A$ and $\mu(A_n) < \infty$ for some n, then show that $\mu(A_n) \downarrow \mu(A)$.
   (c) Show that the second conclusion may fail if $\mu(A_n) = \infty$ for all n.